

# VAPO

## Value-Model-Based RL for Advanced Reasoning

**60.4 on AIME 2024 (avg@32)**

↑ 10+ points above DAPO (50) and DeepSeek-R1-Zero (47)

Base model: Qwen2.5-32B | No SFT data | 5,000 training steps

*ByteDance Seed* · [arxiv 2504.05118](#)

# Why Value Models? Finer Credit Assignment for Long Chain-of-Thought

*Value-model-free methods have limitations — here is why value models matter*

## Value-Model-FREE (GRPO / DAPO)

Assigns trajectory reward uniformly  
across ALL tokens in the response

- Coarse credit assignment
- High-variance Monte Carlo estimates
- Cannot isolate step-level errors
- Same advantage for good & bad tokens



## Value-Model-BASED (VAPO)

Learns a value function for token-level  
credit assignment

- Precise per-token advantage
- Lower-variance estimates
- Generalises across samples
- Isolates subtle step-level mistakes

*Critical for long-CoT: subtle step-level errors cascade into wrong final answers — token-level credit assignment is essential.*

# Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

## ① Value Model Bias

Long trajectories + bootstrapped learning  
accumulate compounding estimation errors.

Value estimates diverge from true returns  
as chain-of-thought length grows.

## ② Heterogeneous Lengths

Responses vary from a few tokens to  
thousands — no single bias-variance  
tradeoff fits all lengths.

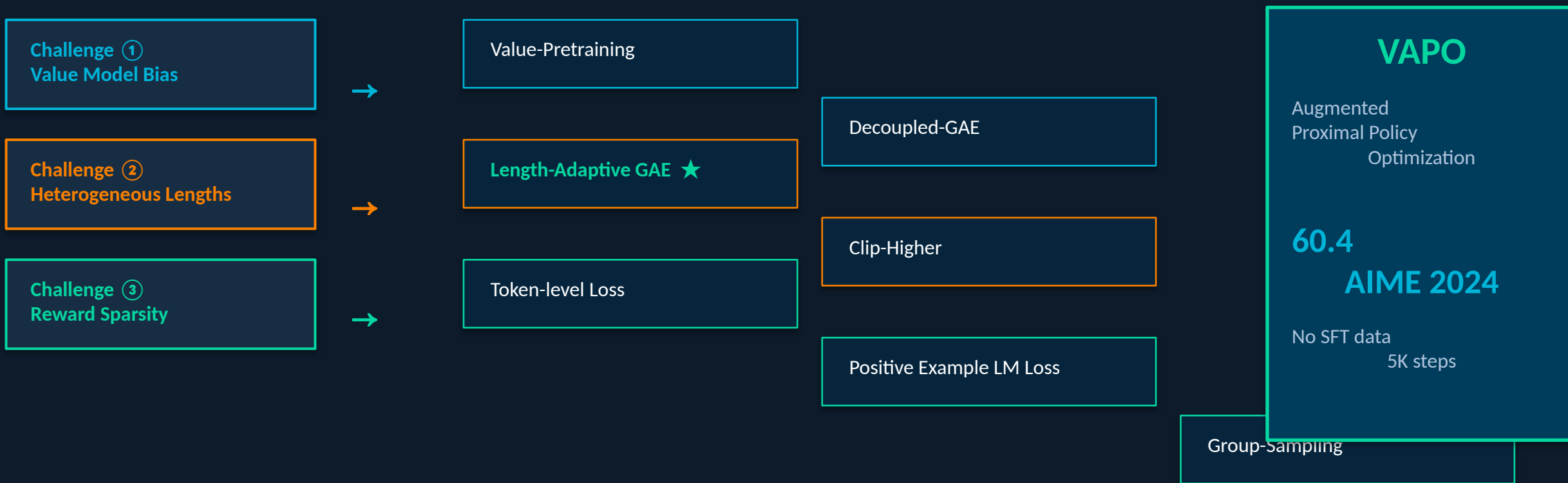
Fixed  $\lambda$  in GAE is suboptimal.

## ③ Reward Signal Sparsity

Only the final <eos> token receives  
a reward signal.

Long chains amplify sparsity, making  
value learning extremely hard.

# VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL



★ *Novel contribution*

# Length-Adaptive GAE: Dynamic $\lambda$ Adjustment for Variable Response Lengths

Novel technique — optimises the bias-variance tradeoff across heterogeneous sequence lengths

## Standard GAE

$$\hat{A}_t = \sum (\gamma\lambda)^k \delta_{\{t+k\}}$$

$\lambda$  fixed for all response lengths

## Problem

Short responses → need low  $\lambda$  (control variance)

Long responses → need high  $\lambda$  (reduce bias)

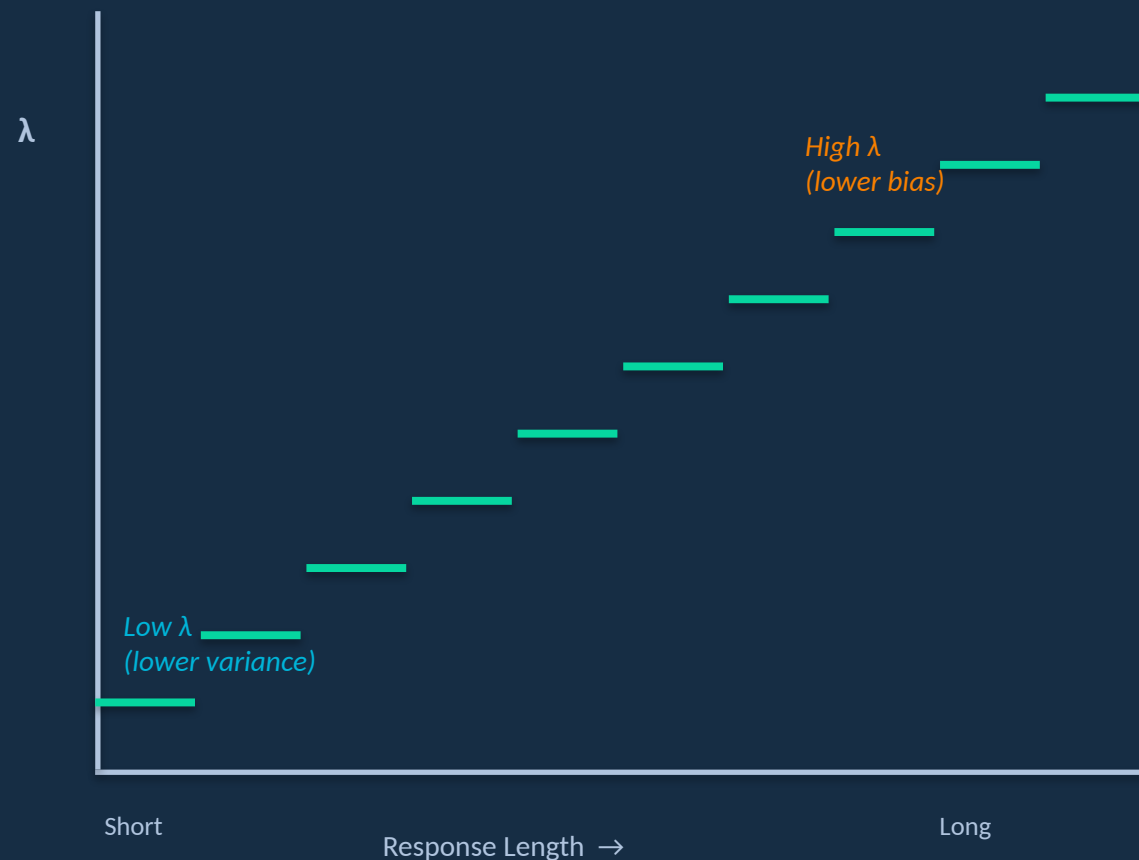
One  $\lambda$  cannot serve both — training degrades.

## Length-Adaptive GAE Solution

$$\lambda(L) = f(\text{response\_length } L)$$

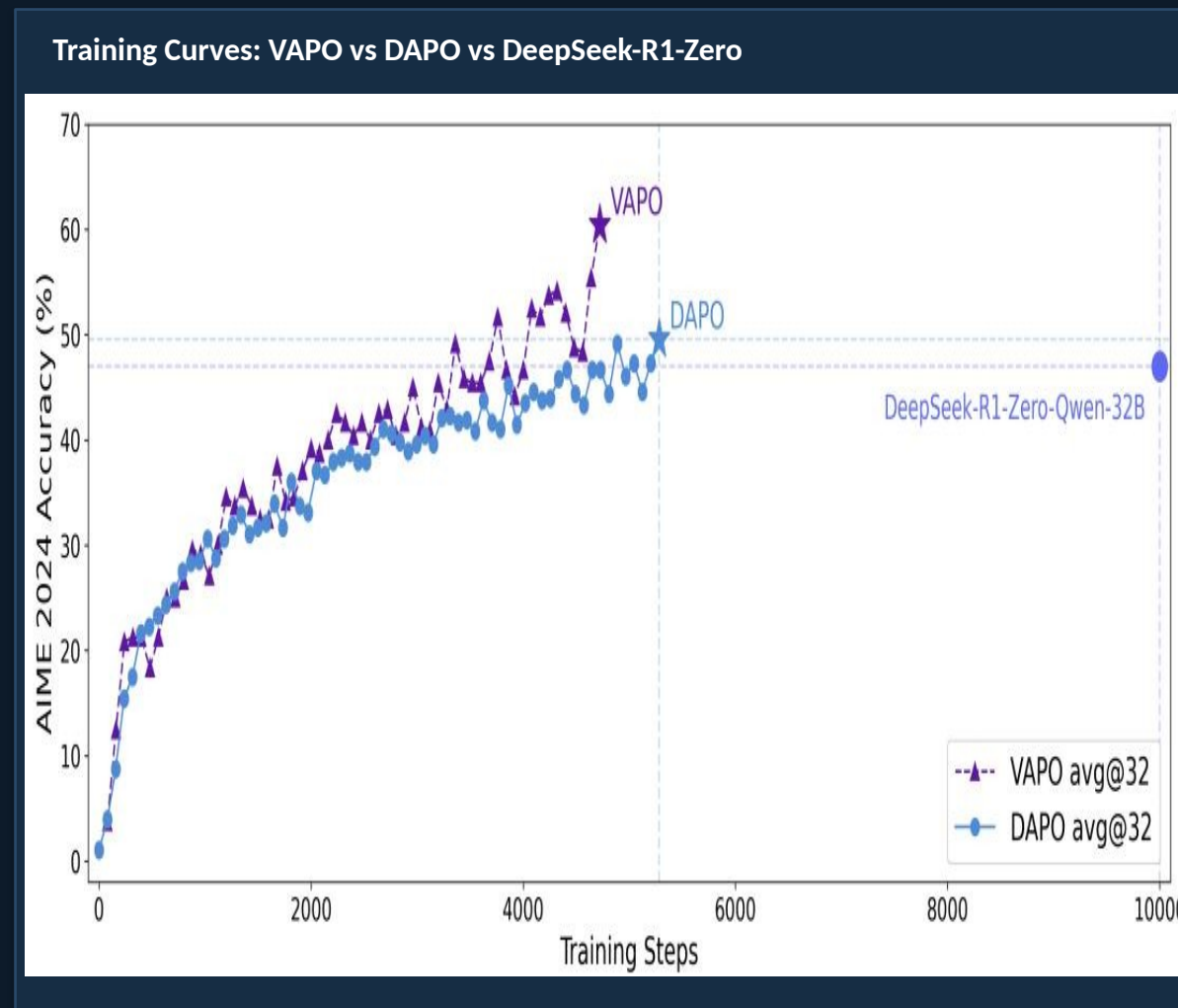
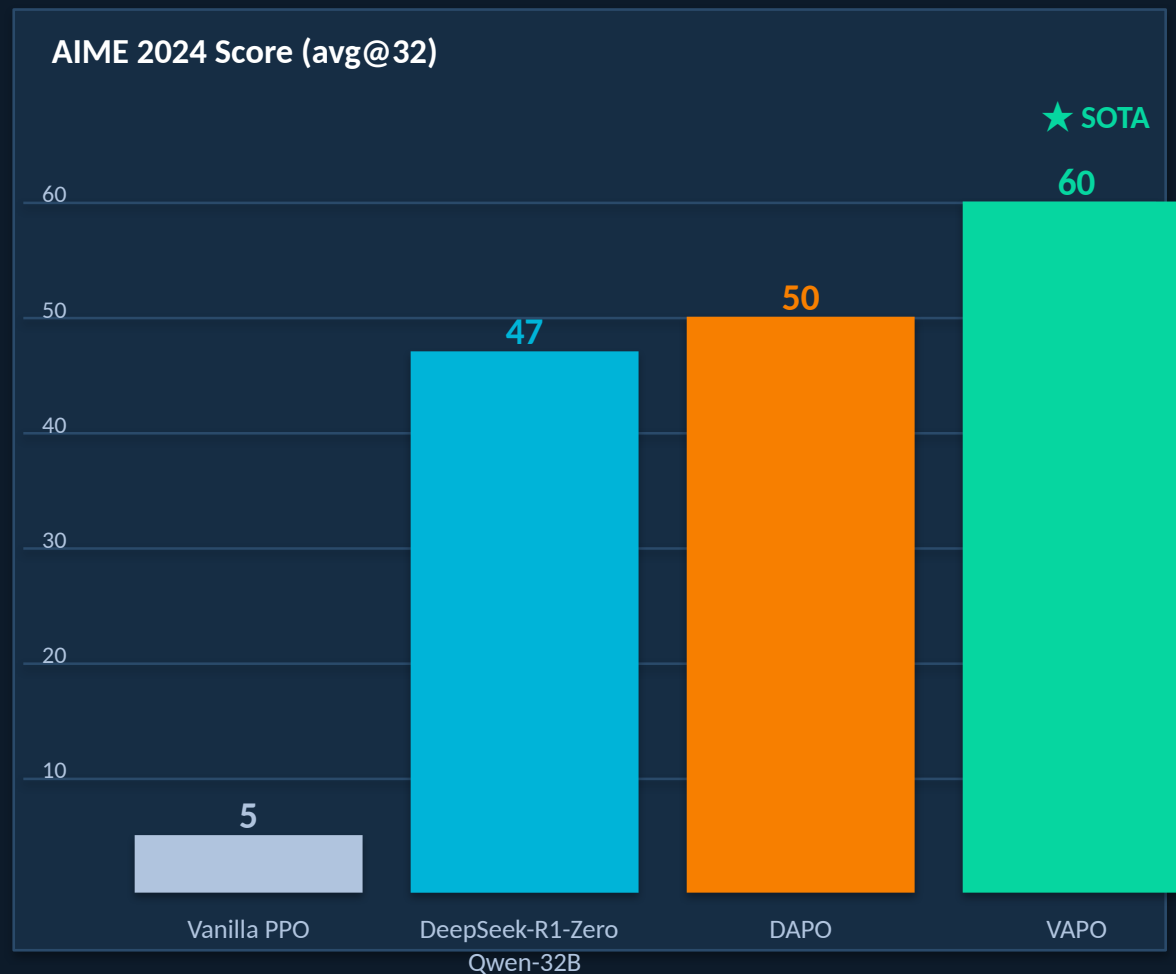
$\lambda$  scales with length → optimal tradeoff at every step

## $\lambda$ vs Response Length



# VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO & DeepSeek-R1-Zero

All models use Qwen2.5-32B base, no SFT data (avg@32)

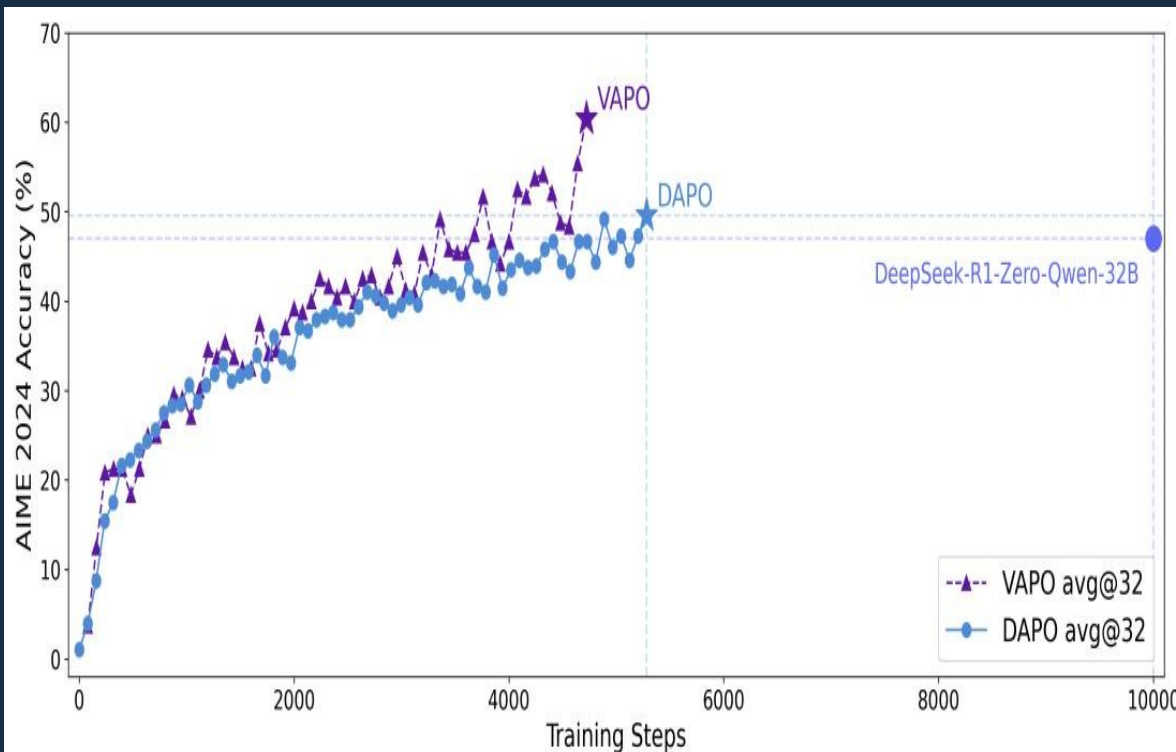


VAPO reaches 60% accuracy at ~4,500 steps; DAPO plateaus at ~50% by 5,500 steps.

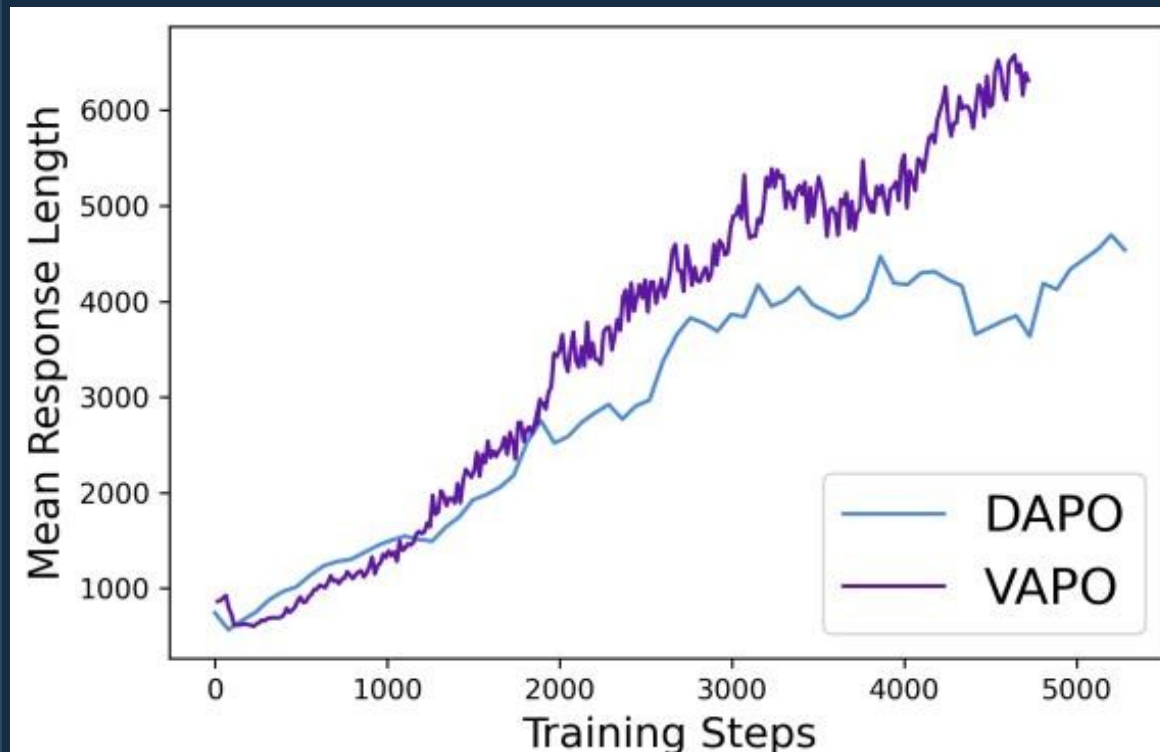
# VAPO Reaches SOTA in 5K Steps — Stable, Efficient, Longer Reasoning Chains

*No training crashes across multiple independent runs*

Accuracy on AIME 2024 over Training Steps



Mean Response Length over Training Steps



- VAPO: ~5,000 steps to SOTA
- VAPO generates ~6,000 token responses vs ~4,500 for DAPO
- DAPO: ~5,500 steps, lower score
- Zero training crashes across multiple runs
- DeepSeek-R1-Zero: ~10,000 steps for 47%

# Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Cumulative ablation from full VAPO (60) removing one component at a time

| Configuration                | AIME Score | Drop |
|------------------------------|------------|------|
| <div></div>                  | 60         | —    |
| w/o Group-Sampling           | 55         | ▼ 5  |
| w/o Positive Example LM Loss | 54         | ▼ 1  |
| w/o Token-level Loss         | 53         | ▼ 1  |
| w/o Clip-Higher              | 46         | ▼ 7  |
| w/o Length-adaptive GAE      | 45         | ▼ 1  |
| w/o Decoupled-GAE            | 33         | ▼ 12 |
| w/o Value-Pretraining        | 11         | ▼ 22 |

⚠ Removing Value-Pretraining collapses performance: 60 → 11 (-49 pts). Without it, value-model-based RL barely functions.



# Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

## 1 Value-model-based RL outperforms value-free methods

When the three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed, VAPO surpasses GRPO/DAPO-style methods by 10+ points.

## 2 Value-Pretraining is the single most critical component

Removing it drops AIME score from 60 → 11 (-49 pts). Pre-training the value model before RL is non-negotiable for stable learning.

## 3 Length-Adaptive GAE solves a fundamental mismatch

Dynamically scaling  $\lambda$  with response length contributes 15 points. Fixed- $\lambda$  GAE is provably suboptimal for heterogeneous-length reasoning tasks.

## 4 VAPO is stable and efficient

60.4 AIME score in ~5,000 steps; zero training crashes across multiple independent runs — a strong signal of robustness for production-scale reasoning model training.

## 5 Implications for future reasoning model development

Value models are a promising backbone for long-CoT RL. Engineering the value model carefully — pretraining, adaptive GAE, decoupled estimation — unlocks performance gains that value-free methods leave on the table.